SYNTHETIC TEST DATA ? NOT AN OFFICIAL AUTOSAR DOCUMENT
Document ID: AUTOSAR-HLD-SYNTH-2026-V2 | Revision: 2.0 | Status: Approved Engineering Baseline
Title: High-Level Design (HLD) Document ? Automotive Powertrain & Body Domain Controller

Section 1: Executive Summary & Architectural Scope

This document specifies the updated High-Level Design (HLD) Revision 2.0 for the Electric Vehicle Powertrain Domain Controller (PDC) and Body Control Module (BCM). Standard AUTOSAR Classic Platform Release 4.4.0 layered model is followed.

1.1 Purpose and Revision 2.0 Highlights

Revision 2.0 introduces Hardware Security Module (HSM) cybersecurity integration, expands battery fast-charging support, aligns signal timing discrepancies, and removes legacy OBD diagnostics in favor of UDS over Ethernet (DoIP).

SYNTHETIC TEST DATA ? NOT AN OFFICIAL AUTOSAR DOCUMENT
Document ID: AUTOSAR-HLD-SYNTH-2026-V2 | Revision: 2.0
Title: High-Level Design (HLD) Document ? Automotive Powertrain & Body Domain Controller

Section 2: Architectural Overview & AUTOSAR Layered Model
The architecture decomposes vehicle functions into distinct, decoupled software components (SWCs).
Task scheduling and memory protection partitions are updated to incorporate dedicated crypto execution contexts on Core 1.

Section 3: Software Components (SWCs) and Port Definitions
The Application Layer defines six primary software components:

3.1 ThrottleControl_SWC
- Provided Ports: PpThrottlePos (interface If_ThrottlePosition).
- Required Ports: RpPedalSensor.

3.2 BrakeManager_SWC
- Provided Ports: PpBrakeTorqueRequest (interface If_BrakeTorque).
- Required Ports: RpWheelSpeed.

3.3 BatteryManagement_SWC
- Provided Ports:
  - PpBMSStatus: Broadcasts battery pack state via interface If_BatteryState.
  - PpFastChargeControl: Manages DC Fast Charging contactors and 800V pre-charge state machine.
- Required Ports: RpCellVoltage.

3.4 ThermalControl_SWC
- Provided Ports: PpFanSpeedCommand.
- Required Ports: RpCoolantTemp (interface If_ThermalData).

3.5 Gateway_SWC
- Central gateway routing between CAN-FD and Automotive Ethernet Backbone.

3.6 SecurityManager_SWC
- Component Type: Security Service Software Component (ASIL-B).
- Description: Manages secure boot authentication, message authentication codes (SecOC), and cryptographic key provisioning.
- Provided Ports:
  - PpCryptoKeyService: Provides cryptographic verification services to application SWCs.

SYNTHETIC TEST DATA ? NOT AN OFFICIAL AUTOSAR DOCUMENT
Document ID: AUTOSAR-HLD-SYNTH-2026-V2 | Revision: 2.0
Title: High-Level Design (HLD) Document ? Automotive Powertrain & Body Domain Controller

Section 4: Software Interfaces and Signal Dictionary
All communications use standardized AUTOSAR Sender-Receiver interfaces.

4.1 Interface Definitions:
- Interface If_ThrottlePosition:
  - Sig_ThrottleAngle: float32, range [0.0, 100.0] %, Update Rate: 10ms (Consistent across all layers).
  - Sig_ThrottleFaultStatus: uint8, range [0, 3], Update Rate: 10ms.
- Interface If_BrakeTorque:
  - Sig_RequestedTorque: int16, range [-500, 500] Nm, Update Rate: 10ms.
  - Sig_BrakePedalState: boolean, Update Rate: 10ms.
- Interface If_BatteryState:
  - Sig_StateOfCharge: uint8, range [0, 100] %, Update Rate: 100ms.
  - Sig_PackVoltage: float32, range [0.0, 800.0] V, Update Rate: 50ms.
  - Sig_PackCurrent: float32, range [-400.0, 400.0] A, Update Rate: 20ms.
- Interface If_ThermalData:
  - Sig_CoolantTemp: int16, range [-40, 150] degC, Update Rate: 500ms.
  - Sig_PumpSpeedTarget: uint16, range [0, 6000] RPM, Update Rate: 100ms.
- Interface If_SecurityVerification:
  - Sig_SecOCAuthStatus: uint8, range [0, 1], Update Rate: 10ms.

SYNTHETIC TEST DATA ? NOT AN OFFICIAL AUTOSAR DOCUMENT
Document ID: AUTOSAR-HLD-SYNTH-2026-V2 | Revision: 2.0
Title: High-Level Design (HLD) Document ? Automotive Powertrain & Body Domain Controller

Section 5: Functional Flows and Inter-SWC Communication
Functional execution chains updated for cybersecurity verification and DC fast-charge coordination.
Flow 1: Driver Acceleration Request Flow (Verified end-to-end within 10ms).
Flow 2: Regenerative Braking Coordination Flow.
Flow 3: Thermal Protection Derating Flow.

SYNTHETIC TEST DATA ? NOT AN OFFICIAL AUTOSAR DOCUMENT
Document ID: AUTOSAR-HLD-SYNTH-2026-V2 | Revision: 2.0
Title: High-Level Design (HLD) Document ? Automotive Powertrain & Body Domain Controller

Section 6: Dependencies, Timing & RTE Interactions
6.1 Timing Synchronization
Transmission period for Sig_ThrottleAngle is unified at 10ms cyclic execution across COM, RTE, and CAN-FD channels, resolving previous ambiguity.

Section 7: Cyber Security Hardware Security Module (HSM) Integration
7.1 Security Architecture
Integrates ISO/SAE 21434 automotive cyber security management system requirements.
Hardware Cryptographic Accelerator handles AES-128 CMAC calculation for all safety-critical CAN-FD frames.
Key storage is securely partitioned within internal HSM secure flash memory.